

WASP-52b

R-0511

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_180807 · created 2026-07-20T20:24:44+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458337.8708 +/- 0.0043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.062
Transit depth (Rp/Rs)^2	3.6 +/- 2.37 [%]
Orbital Inclination (inc)	83.37 +/- 2.8
Airmass coefficient 1 (a1)	2.5 +/- 1.15
Airmass coefficient 2 (a2)	-0.74 +/- 1.38
Scatter in the residuals of the lightcurve fit is	47.59 %
Best Comparison Star	#2 - [218, 217]
Optimal Aperture	1.69
Optimal Annulus	8.05
Transit Duration (day)	0.037 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.19 ± 0.062 vs 0.1646 (deviation 0.41σ)
Duration fit vs published	0.037 d vs 0.0758 d (deviation 1.94σ)
Mid-time O–C	10.5 min
Depth SNR	3.1
Residual scatter	47.59 %

Light curve

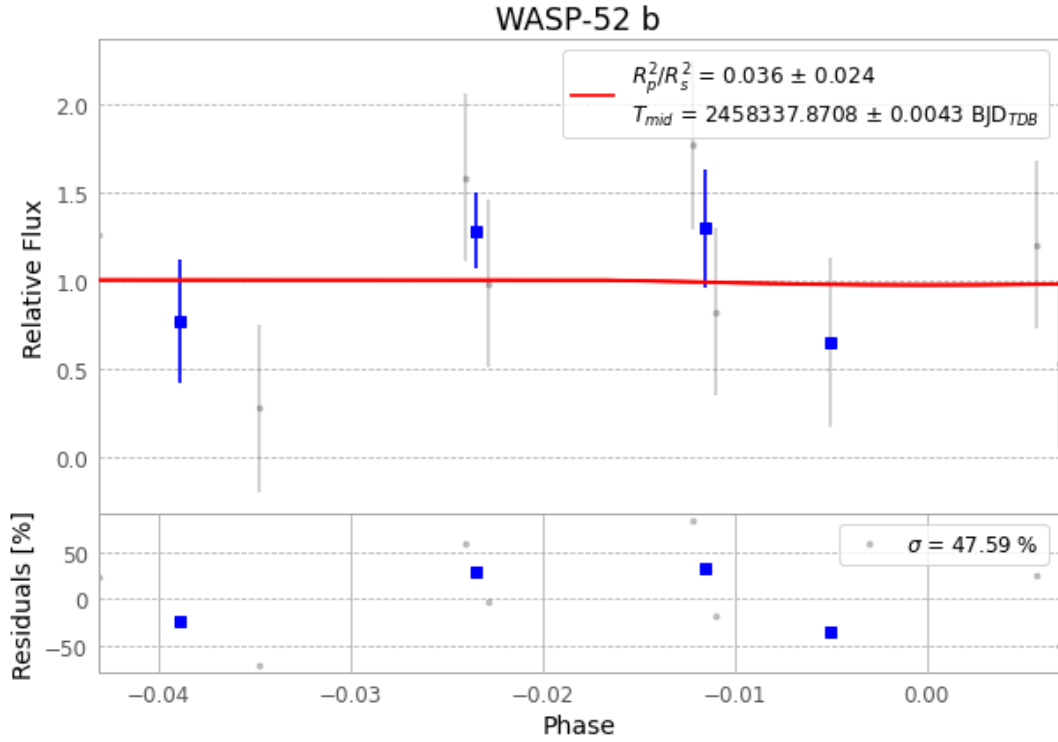


Figure 1 — FinalLightCurve_WASP-52 b_2018-08-06.png (SHA-256 0ebddce917598ae4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [359, 81], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f36deb3fe3362117...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

77 FITS frames (51 MB), WASP-52180807064812.FITS → WASP-52180807103612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-180807.log (0d0fbc6239a7...), FinalLightCurve_WASP-52 b_2018-08-06.png (0ebddce91759...), FinalLightCurve_WASP-52 b_2018-08-06.csv (6318b024bb4b...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 20:24Z.